1] What is simple linear regression?

A technique that models the relationship between one independent variable (X) & one dependent variable (Y) using a straight line.

Eqn : $Y = b_0 + b_1 X$

2] What is Multiple linear regression?

A regression with two or more independent variables

Eqn: $Y = b_0 + b_1 X_1 + b_2 X_2 + \dots + b_n X_n$

3] <u>Mathematical implementation of linear regression</u>

(1) $y = \beta_0 + \beta_1 x$

where β_0 = Intercept (value of y when $x = 0$)

β_1 = slope (how much y changes when x changes)

(2) Objective (Minimize error) : Find best line by ↓ MSE

$$MSE = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

where : y_i - actual value

$\hat{y}_i$ - predicted value

(3) Formula for coefficients

Slope :

$$\beta_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

Intercept :

$$\beta_0 = \bar{y} - \beta_1 \bar{x}$$

Example

Data:-

x (Hours studied)	y (Marks)
1	2
2	4
3	5
4	4
5	5

step 1 : compute Means :- $\bar{x} = 3$ $\bar{y} = 4$

step 2 : compute slope β_1

$$\beta_1 = \frac{(1-3)(2-4) + (2-3)(4-4) + (3-3)(5-4) + (4-3)(4-4) + (5-3)(5-4)}{(1-3)^2 + (2-3)^2 + (3-3)^2 + (4-3)^2 + (5-3)^2}$$

$$= \frac{6}{10} = 0.6$$

step 3 : compute Intercept β_0 : $\beta_0 = 4 - (0.6 \times 3) = 4 - 1.8$

$= 2.2$

Final model,

$$y = 2.2 + 0.6x$$

step 4 : Prediction Example

If a student studies 6 hours

$$y = 2.2 + 0.6(6) = 2.2 + 3.6$$

$$y = 5.8$$

Predicted marks = 5.8 //

4] Mathematical implementation of multi linear regression

(i) Model Eq'n : $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$

y - dependent var

$x_1, x_2, \dots, x_n$ - independent var

β_0 - intercept

$\beta_1, \beta_2, \dots, \beta_n$ - coeffecients

(ii) Matrix Form : $Y = X\beta + \epsilon$

Y - output vector

X - feature matrix

β - coeffecient vector

ϵ - error

(iii) Optimal sol'n (Normal Eq'n):

$$\beta = (X^T X)^{-1} X^T Y$$

Example: Problem : Predict marks (y) based on :

x_1 = Hours studied

x_2 = Hours slept

Data:

Hours studied (x_1)	Sleep (x_2)	Marks (y)
1	6	2
2	5	3
3	6	5
4	5	6

Step 1: Form Matrix X , (Add column of 1s for intercept)

$$X = \begin{bmatrix} 1 & 1 & 6 \\ 1 & 2 & 5 \\ 1 & 3 & 6 \\ 1 & 4 & 5 \end{bmatrix} \qquad Y = \begin{bmatrix} 2 \\ 3 \\ 5 \\ 6 \end{bmatrix}$$

Step 2: Apply formula , $\beta = (X^T X)^{-1} X^T Y$

$\beta_0 = -1, \quad \beta_1 = 1.5, \quad \beta_2 = 0.5$

Final model , $y = -1 + 1.5 x_1 + 0.5 x_2$

Step 3 : Prediction. For ex, study = 5, slep = 6

$y = -1 + 1.5(5) + 0.5(6) = 9.5$ //

5] Loss function:

A function that measures how wrong the model predictions are

6] Polynomial regression:

A regression where the relationship is curved instead of linear.

$$Y = b_0 + b_1 X + b_2 X^2 + \dots + b_n X^n$$

7] Optimization:

The process of finding best model parameters (like weights) by minimizing loss

8] Evaluation metrics for regression:

- Mean squared Error (MSE)
- Mean Absolute Error (MAE)
- Root mean squared Error (RMSE)
- R^2 score

9] Gradient Descent,

An optimization algorithm used to minimize loss by updating parameters step-by-step.

$$\theta = \theta - \alpha \frac{\partial J}{\partial \theta}$$

10] Assumptions of Linear Regression :

- Linear relationship
- No multicollinearity
- constant variance
- Independence of errors
- Normal distribution of errors

11] Overfitting,

Model learns too much (including noise) → performs well on training, poorly on test data

12] Underfitting :

Model is too simple – cannot capture patterns

poor performance everywhere

13] How to take the degree in polynomial regression?

- Start small degree (2 or 3)
- Use validation / testing
- Choose degree with best performance (not over

14) <u>Mean Square Error (MSE):</u>

$$MSE = \frac{1}{n} \sum (y_i - \hat{y_i})^2$$

Measures average squared error

15) <u>Mean Absolute Error (MAE):</u>

$$MAE = \frac{1}{n} \sum |y_i - \hat{y_i}|$$

Measures average absolute error

16) <u>Root Mean Squared Error (RMSE):</u>

$$RMSE = \sqrt{MSE}$$

Gives error in same unit as output

17) What is Lasso Regression?

Uses L1 regularization

Adds penalty: $\lambda \sum |b_i|$

It is a linear regression technique that adds a penalty equal to the absolute value of coeffecients to reduce overfitting.

Mathematical form: $L_{ou} = \underbrace{\sum (y_i - \hat{y}_i)^2}_{MSE} + \underbrace{\lambda \sum |\beta_j|}_{\text{penalty (L1 reg)}}$

λ - controls strength of penalty

Why use Lasso?

- Helps reduce overfitting
- Automatically sets some coeffecients to zero
- Performs feature selection

When to Use Lasso:

- You have many features
- Some features are not important
- You want simpler, interpretable model
- Dataset may contain noise or irrelevant variables

18] Ridge regression :

Ridge regression (L2 regularization) adds a penalty equal to the square of coefficients to reduce overfitting.

Mathematical form : $Loss = \sum (y_i - \hat{y}_i)^2 + \lambda \sum \beta_j^2$

Advantages :

- Reduces overfitting
- Handles multicollinearity
- Keeps All Features
- Improves Model stability